Partial formulation of an ammonia volatilization model

State variable = "total ammoniaical nitrogen" = ammonia as N = mass of sum of NH_3 and NH_4^+

Mass balance

Ammonia accumulation = Ammonia pumped into tank - Ammonia volatilized out ~~+ generation~~ - Ammonia pumped out

with no urea hydrolysis

Let's assume the tank has a constant volume and continuous pumping rates

Constitutive equations

$$\dot{m}_{in} = c_0 \cdot \dot{V}$$

$\dot{V}$ — pumping rate ($m^3 s^{-1}$)

c_0 — conc. of ammonia N in fresh slurry ($g\ m^{-3}$)

$$\dot{m}_{out} = c \cdot \dot{V}$$

$$J = K_L (c_{NH_3} - c^*_{NH_3,air})$$

c_{NH_3} — Bulk $NH_3(aq)$ conc. in slurry. ($g\ m^{-3}$)

$c^*_{NH_3,air}$ — Background conc. in air, converted to liquid phase units. ($g\ m^{-3}$)

Note that J depends on free NH3 concentration, not total ammonia. So we need a linkage equation.

Linkage equation

From mass action expression and mass balance . . .

$$\frac{c_{NH_3} \cdot c_{H^+}}{c_{NH_4^+}} = K_a \qquad c_{TAN} = c_{NH_4^+} + c_{NH_3}$$

You can derive this handy equation

$$\boxed{c_{NH_3} = \frac{c_{TAN}}{1 + \underbrace{c_{H^+}}_{\approx 10^{-pH}}/K_a}}$$

$NH_4^+ \rightleftharpoons NH_{3(aq)} + H^+$

is an extremely fast proton exchange reaction, so we will assume equilibrium at all times and locations

There are various forms that you can get to by rearranging

And we could consider this a linkage equation too

$$c^*_{NH_3,air} = c_{NH_3,air} \cdot K_H$$

So, governing equation

$$\frac{dm}{dt} = \dot{V} \cdot c_{TAN,0} - A \cdot k_L \cdot \left(\frac{c_{TAN}}{1 + 10^{-pH}/K_a} - c^*_{NH_3,air}\right) - \dot{V} \cdot c_{TAN}$$

Divide by slurry volume

$$\frac{dc_{TAN}}{dt} = \underbrace{\frac{c_{TAN,0}}{\tau}}_{\text{Retention time}} - \underbrace{\frac{k_L}{d}}_{\text{Depth} = \frac{A}{V}} \left(\frac{c_{TAN}}{1 + 10^{-pH}/K_a} - c^*_{NH_3,air}\right) - \frac{c_{TAN}}{\tau}$$

We can put either of these directly in a `rates()` function for a numeric solution.

For an analytical or steady-state solution, we need to simplify.

$$\frac{dC_{TAN}}{dt} = \frac{C_{TAN,0}}{\tau} - \frac{k_L}{d}\left(\frac{C_{TAN}}{1+10^{-pH}/k_a} - C^*_{NH_3,air}\right) - \frac{C_{TAN}}{\tau}$$

$$0 =$$

$$\frac{(C_{TAN} - C_{TAN_0})}{\tau} = -\frac{k_L}{d}\left(\frac{C_{TAN}}{1+10^{-pH}/k_a} - C^*_{NH_3,air}\right)$$

$$C_{TAN} - C_{TAN_0} = -\underbrace{\frac{\tau\cdot k_L}{d\cdot(1+10^{-pH}/k_a)}}_{\equiv K}\cdot C_{TAN} - \underbrace{\frac{\tau\cdot k_L\cdot C^*_{NH_3}}{d}}_{\equiv B}$$

$$C_{TAN} - C_{TAN_0} = -K\cdot C_{TAN} - B$$

$$(1+K)\cdot C_{TAN} = C_{TAN,0} - B$$

$$C_{TAN} = \frac{C_{TAN,0} - B}{1+K}$$ at steady state